



Long-Term Cognitive Improvement After Functional Hemispherectomy

Xiao-Peng Qu, Yan Qu, Chao Wang, Bei Liu

■ **BACKGROUND:** To investigate intellectual development after functional hemispherectomy and to evaluate the favorable or unfavorable factors affecting prognosis.

■ **METHODS:** Twenty-three patients from the patient follow-up database of the Epilepsy Center of Neurosurgery were selected for this retrospective study. The inclusion criteria of the patients were functional hemispherectomy surgery, surgical age older than 6 years, patients able to complete the intelligence assessment, follow-up time not less than 2 years, and availability of complete preoperative evaluation and postoperative follow-up data. Full Scale Intelligence Quotients (FSIQ) were calculated preoperatively and at the follow-up times for each patient. Correlations between the preoperative to postoperative changes in FSIQ and risk factors (age at epilepsy onset, duration of preoperative epilepsy, postoperative changes in the electroencephalogram, seizure status after surgery, and follow-up time) were analyzed. EpiData 3.1 software was used to establish a database and SPSS 22.0 software was used for data collation and analysis, and used the matching sample Wilcoxon symbol rank and test to conduct the analysis.

■ **RESULTS:** The data of 23 patients were analyzed. The final follow-up results showed that 73.9% of the patients (17/23) had a significant increase in FSIQ scores ($P = 0.006$) after surgery, indicating that their cognitive function improved significantly. Successful surgery, successful postoperative seizure control, normalization of postoperative electroencephalogram, shorter disease duration before surgery, and longer recovery periods after surgery (follow-up time) were all related with good intellectual prognosis.

■ **CONCLUSIONS:** Successful functional hemispherectomy significantly improved the cognitive function of patients. Functional hemispherectomy is a safe and effective treatment for patients with intractable epilepsy caused by hemispheric lesions and can effectively promote intellectual recovery and development.

INTRODUCTION

Functional hemispherectomy, resulting from hemispheric resection, can maximize the severance of white brain matter by the minimal removal of brain tissue accomplished by isolating the epileptogenic focus, destroying the connection between the epileptogenic focus and the residual functional hemisphere, and blocking the conduction pathway of epileptic discharge. This procedure effectively avoids the possible complications of brain displacement, hydrocephalus, and hemosiderin deposition on the surface of the brain after extensive hemispherectomy. For patients with intractable epilepsy originating from one side of the cerebral hemisphere who have a relatively well-functioning contralateral hemisphere, functional hemispherectomy can achieve better seizure control and significantly improve the patient's quality of life. The postoperative complete remission rate of epilepsy following hemispherectomy has been reported to range from 70% to 90%.¹⁻⁴ As the symptoms of the seizures were alleviated, the patient's cognition, especially language function, significantly improved. However, these reports focused on anatomical hemispherectomies.⁵⁻¹⁰

The factors contributing to cognitive dysfunction include the major causes of epilepsy, seizures themselves, interictal epileptiform discharges, and the adverse effects of antiepileptic drugs.¹¹ For patients with surgically curable epilepsy, successful surgery

Key words

- Cognitive improvement
- Epilepsy surgery
- FSIQ
- Functional hemispherectomy

Abbreviations and Acronyms

EEG: Electroencephalogram
FSIQ: Full-Scale Intelligence Quotients
IQ: Intelligence quotient

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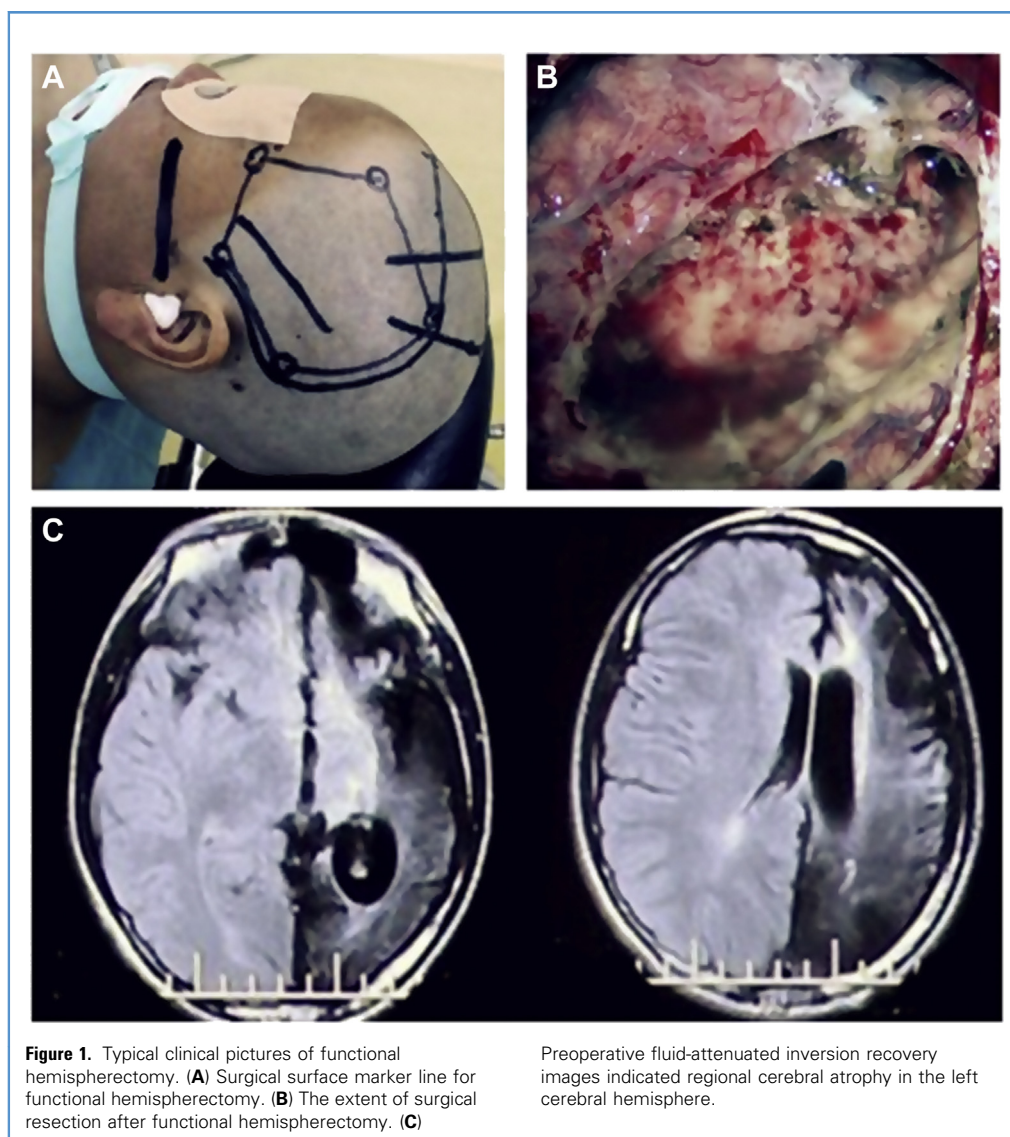
Xiao-Peng Qu and Yan Qu contributed equally to this article.

Citation: World Neurosurg. (2020) 135:e520-e526.
<https://doi.org/10.1016/j.wneu.2019.12.058>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

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may affect all of these factors¹²⁻¹⁵ (such as the major causes of epilepsy, seizures themselves, interictal epileptiform discharges, and the adverse effects of antiepileptic drugs). Studies have associated hemispherectomy surgery with greater cure rates and fewer complications, but few studies have been conducted on the effect on postoperative patient cognition. This may be because the postoperative follow-up times were short and most children who undergo this surgery are younger and have lower levels of cognitive development, making it difficult to conduct standardized cognitive assessments.¹⁶

The primary objective of this study was to investigate the effects of functional hemispherectomy on long-term cognitive development. A secondary objective was to analyze the correlation between multiple prognostic risk factors and cognitive function development (e.g., epilepsy duration, age at surgery, and seizures). At long-term follow-up, functional hemispherectomy was

found to have positive effects on recovery and the development of patients' intelligence, while controlling seizures. Positive surgical results may make the patients and their families more active and willing to undergo rehabilitative therapy, which will further affect the patient's behavior, communication, and even learning, other life skills, and employment.

METHODS

Patient Population and Data Collection

Since 2000, a total of 263 patients have undergone preoperative evaluations and epilepsy surgery at our research center, of whom 61 patients underwent functional hemispherectomies. Patient data were screened to determine: 1) that the type of surgery was functional hemispherectomy, 2) the age at surgery was more than 6 years, 3) the surgery had been completed for at least 2 years, and

Table 1. Classification by Wieser et al.¹⁸

Class	
1a	Completely seizure-free since surgery, no auras
1	Completely seizure-free, no auras within the last evaluated year*
2	Only auras, no other seizures
3	1–3 seizures per year $\pm$ auras
4	4 seizure days† per year to 50% reduction of baseline seizure days‡
5	<50% reduction of baseline seizure days to 100% increase of baseline seizure days $\pm$ auras
6	>100% increase of baseline seizure days $\pm$ auras
Seizure days during the first month after surgery are not counted.	
*Seizure outcome class is determined for each year at annual intervals after surgery.	
Patients may change from one class to another from year to year.	
†A seizure day is a 24-hour period with 1 or more seizures. This may include an episode of status epilepticus.	
‡The number of baseline seizure days is calculated by determining the seizure-day frequency during the 12 months before surgery.	

4) complete preoperative evaluation and postoperative follow-up data were available. In the end, 23 patients were included in the study, including pediatric patients 16 years old or younger (14/23) and adult patients 17 years old and older (9/23). Thirty-eight cases did not meet the following inclusion criteria: 24 patients were 6 years old or younger at the time of surgery and had inaccurate cognitive assessments, the follow-up time in 8 patients was less than 2 years, and the follow-up data in 6 patients were insufficient.

Preoperative evaluations followed the standard protocol,¹⁷ which included neurologic, ophthalmologic, and neuropsychologic assessment, as well as enhanced video electroencephalogram (EEG) monitoring, fluorodeoxyglucose–positron emission tomography, and high-resolution magnetic resonance imaging. All surgeries were performed by one neurosurgeon using the same technique (Figure 1). Postoperative follow-ups were scheduled for 3 months after surgery and once a year after surgery. The follow-up examinations included complete neuropsychological assessments, 24-hour EEG, and magnetic resonance imaging examinations. In addition, data were collected on seizure etiology, the age at onset of seizures, age at surgery, type of epilepsy, type of surgery, performance, and motor function of acute postoperative seizures. The seizure outcomes were classified according to the International League Against Epilepsy criteria¹⁸ (Table 1) because the criteria are more complete and overcome several disadvantages of the Engel outcome classification widely used at present.¹⁹

The acquisition of these data was approved by Tangdu Hospital Affiliated to Fourth Military Medical University, and the patients and/or their caregivers signed informed consent in accordance with the standards of the Declaration of Helsinki.

Each participant was individually tested by a professional researcher. The children group (14/23) and the adult group (9/23) were tested by the Wechsler Intelligence Scale for Children and the Wechsler Adult Intelligence Scale, respectively. Higher-order assessment indices, namely the General Ability Index and the

Cognitive Productivity Index, were used to analyze the patients' overall intelligence levels and obtain overall intelligence quotient (IQ) values (Full Scale Intelligence Quotients [FSIQ]). The cognitive tests were scored according to the following criteria: FSIQ ≤ 69 was scored as mental deficiency; FSIQ 70–79, as borderline intellectual disability; FSIQ 80–89, as mild cognitive delay; and FSIQ 90–100, as normal.^{20,21} To investigate the effects of functional hemispherectomy on cognition and the possible risk factors associated with cognition, possible links between the preoperative FSIQ values and the postoperative FSIQ values at the last follow-up and the major clinical parameters with IQ pre/postoperative changes were analyzed. The results are shown in Table 2.

Statistical Analyses

The database was built by EpiData 3.1 software (<https://www.epidata.dk/download.php>), and the data were sorted and analyzed by SPSS 22.0 software (IBM Corp., Armonk, New York, USA). The measurement data are expressed as means and standard deviation ($\bar{x} \pm s$), and the enumeration data are expressed as frequencies (%). The FSIQ differences before and after the surgery were analyzed by the paired sample Wilcoxon signed-rank sum test. All statistical analyses were conducted by bilateral tests, and $P < 0.05$ was statistically significant.

RESULTS

Description of the Study Group and Patient Characteristics

The mean postoperative follow-up time for 23 patients (11 females) meeting the study criteria was 5.6 ± 2.5 years (range: 2–10 years). The mean age at epilepsy onset was 6.1 ± 5.4 years (range: 0.2–19 years) and 43.5% of the patients (10/23) developed seizures before the age of 7 years. The mean age at hemispherectomy surgery was 15.7 ± 6.1 years (range: 6.5–28.0 years) and 60.9% of the patients were minors younger than 16 years old at the time of surgery (14/23). During the last follow-up of patients who underwent hemispherectomy, 70% (16/23) of the patients were seizure-free according to International League Against Epilepsy criteria 1a/1.

The mean duration of preoperative seizures was 9.57 ± 6.8 years (range: 0.4–27.4 years) and 82.6% of the patients (19/23) had had seizures for more than 2 years. The EEG data showed bilateral epileptic seizure-like discharges in 39.1% of the preoperative patients (9/23) and contralateral hemisphere epileptic discharges were still present in 47.8% of the postoperative patients (11/23).

Pre- to Postsurgical Gains and Losses in FSIQ

The detailed changes in FSIQ of the patients before and after functional hemispherectomy are listed in Table 2. The mean preoperative FSIQ score was 77.43 ± 9.179 (range: 61–92). Preoperative intelligence developmental delay (FSIQ ≤ 89) was found in 91.3% of the patients (21/23), which included 6 patients with mental deficiency, 7 patients with borderline intellectual disability, and 8 patients with mild delay. The test results showed that only 2 patients had normal cognitive development. The mean value of the FSIQ scores at the last follow-up after surgery was 83.30 ± 13.529 (range: 57–100). The postoperative cognitive assessment results in 13 patients were normal (56.5%, 13/23). The test results showed mental deficiencies in 5 patients, 2

Table 2. Demographic Data of the Study Group

Patient/ Sex	Epilepsy Onset	Duration of Epilepsy	Age at Surgery	Pathology	Follow up	Side of Surgery	Lateralization of Spikes, Preoperative	Spikes in the Contralateral Hemisphere, Postoperative	Seizure Baseline	Seizure Type	Classification, Postoperative	FISQ Preoperative	FISQ, Postoperative
1/F	19	0.6	20	Trauma	3	Right	Right	0	1–3/ month	CPS	1a	91	93
2/M	10	12	22	Stroke	4	Left	Left	0	1–3/ week	SPS+CPS+SG	1a	82	91
3/F	8	5	13	FCD	4	Right	Bilateral	0	1–3/ week	CPS+SG	1a	68	83
4/F	9	8	17	RE	4	Right	Bilateral	0	4–10/ month	CPS	1a	74	81
5/F	12	13	25	FCD	5	Left	Bilateral	0	1–3/ week	CPS+SG	1a	82	90
6/F	0.6	27.4	28	Stroke	6	Right	Right	1	1–3/ month	CPS+SG	1	76	84
7/M	0.4	11.6	12	FCD	6	Right	Right	1	1–3/day	SPS+CPS+SG	5	64	59
8/M	1	22	23	PMG	8	Left	Left	1	1–3/ month	CPS+SG	5	61	57
9/F	9	5	14	Stroke	5	Right	Bilateral	1	1–3/ week	CPS+SG	1a	86	95
10/M	13	8	21	RE	2	Left	Left	1	1–3/day	SPS+CPS+SG	3	89	90
11/M	7	18	25	Trauma	4	Right	Bilateral	1	1–3/ week	CPS	1	84	90
12/M	1	5.5	6.5	MCD	3	Left	Bilateral	1	1–3/day	CPS+SG	2	79	76
13/M	4	8	12	Trauma	7	Left	Left	0	1–3/day	CPS	1a	77	96
14/M	1	12	13	PMG	6	Right	Right	1	>10/ month	CPS+SG	5	68	60
15/M	7	0.4	7.4	Stroke	7	Left	Bilateral	0	4–10/day	CPS	1a	92	100
16/M	12	6	18	SWS	7	Right	Right	1	1–3/ month	CPS+SG	1	87	91
17/F	13	1	14	Trauma	10	Right	Right	0	4–10/ week	CPS+SG	1a	83	99
18/M	1.5	9	10.5	PMG	10	Left	Bilateral	1	4–10/ week	SPS+CPS+SG	3	75	62

The unit of time is years.

FISQ, Full Scale Intelligence Quotients; F, female; CPS, complex partial seizures; M, male; SPS, simple partial seizures; SG, secondary generalization; FCD, focal cortical dysplasia; RE, Rasmussen encephalitis; PMG, polymicrogyria; MCD, mild cortical dysplasia; SWS, Sturge–Weber syndrome.

Continues

Table 2. Continued

Patient/ Sex	Epilepsy Onset	Duration of Epilepsy	Age at Surgery	Pathology	Follow up	Side of Surgery	Lateralization of Spikes, Preoperative		Spikes in the Contralateral Hemisphere, Postoperative	Seizure Baseline	Seizure Type	Classification, Postoperative		FSIQ, Preoperative	FSIQ, Postoperative
19/F	7.5	0.5	8	FCD	8	Left	Left	Left	0	1–3/day	CPS	1		72	91
20/F	2.5	13.5	16	Stroke	3	Left	Left	Left	0	1–3/day	CPS+SG	1a		86	93
21/F	0.2	11.3	11.5	RE	4	Left	Left	Left	0	>10/ month	CPS	1a		64	77
22/M	0.4	14.6	15	MCD	3	Left	Left	Left	1	1–3/ month	SPS+CPS+SG	3		68	67
23/F	1.3	7.7	9	SWS	9	Right	Bilateral		0	4–10/ week	CPS+SG	1a		73	91

The unit of time is years.

FSIQ, Full Scale Intelligence Quotients; F, female; CPS, complex partial seizures; M, male; SPS, simple partial seizures; SG, secondary generalization; FCD, focal cortical dysplasia; RE, Rasmussen encephalitis; PMG, polymicrogyria; MCD, mild cortical dysplasia; SWS, Sturge–Weber syndrome.

patients with borderline intellectual disability, and 3 patients with mild delay. Normal postoperative cognitive scores were found in 2 cases with normal preoperative FSIQ test results and the scores increased postoperatively. Increases in postoperative FSIQ scores also were seen in 8 patients with mild intellectual disability and in 7 patients with borderline intellectual disability. Normal cognitive levels were found in 3 cases in which the FSIQ scores increased by more than 10 points and decreased postoperative FSIQ scores were found in 2 cases, of which a 13-point decrease was found in a male child with polymicrogyria accompanied by severe intelligence deficits. Two-thirds of the postoperative FSIQ scores of 6 patients with preoperative intelligence deficits were still identified as intellectual disability (66.7%, 4/6).

On the whole, 73.9% of the patients (17/23) had increased postoperative FSIQ scores and improvements in cognition. The differences between the preoperative and postoperative FSIQ scores were statistically significant ($P = 0.006$).

Predictors for Postoperative Changes in FSIQ

As shown in Table 3, patients with an age at epilepsy onset older than 1 year (16/23) had greater postoperative FSIQ scores than those with epilepsy onset at younger ages and the difference was statistically significant ($P = 0.032$). After the hemispherectomy, patients with no postoperative contralateral hemisphere discharge (12/23) had greater increases in postoperative IQ scores, and the difference was statistically significant ($P = 0.001$). The IQ scores of patients free of seizures postoperatively (16/23) were significantly greater than those in patients who were not free of seizures postoperatively ($P < 0.001$).

The IQ scores of patients with preoperative epilepsy durations less than or equal to 5 years (6/23) were compared with those of patients with longer epilepsy courses. The differences in the postoperative increase in the patients were statistically significant ($P = 0.05$). The shorter the preoperative disease course, the greater the increase in postoperative mental development.

There was no significant difference between the FSIQ changes in children ≤ 16 years (14/23) and adults > 17 years (9/23) ($P = 0.377$). Likewise, there were no statistically significant differences between the length of follow-up and changes in cognitive function.

DISCUSSION

Surgery is an important means of treating refractory seizures caused by cerebral hemispheric lesions. The purpose of functional hemispherectomy is to functionally separate and isolate the epileptic regions widely distributed throughout the hemisphere. Multilobar/hemispheric epilepsy surgery technology, whereby the brain affected by severe epilepsy is functionally isolated safely and effectively, is constantly improving.^{5,22} At the same time, functional hemispherectomy avoids the high incidence of late complications caused by anatomical hemispherectomy. Therefore, this article focused on the long-term follow-up of patients after functional hemispherectomies and evaluated their cognitive function changes. Generally, the cognitive function in the patients included in this study was significantly improved after functional hemispherectomy.

Table 3. Predictors for Postoperative Changes in FSIQ

Variables	Mean Rank	Z Value	P Value
Age at epilepsy onset, years		−2.142	0.032
≤1	7.43		
>1	14.00		
Spikes, postoperative		−3.360	0.001
Spike activity	7.05		
No spikes	16.54		
Seizure free, postoperative		−3.748	0.001
Seizure free	15.50		
No seizure free	4.00		
Duration of epilepsy, years		−1.964	0.05
≤5	16.67		
>5	10.35		
Age at surgery, years		−0.883	0.377
≤16	13.00		
>16	10.44		

The data were analyzed by the paired sample Wilcoxon signed-rank sum test.
FSIQ, Full Scale Intelligence Quotients.

Most of the studies on cognitive changes to date have included different surgical approaches for the treatment of epilepsy, and there have been few studies on postoperative cognitive function changes in patients who underwent functional hemispherectomies. Studies^{5,23} have reported that postoperative cognitive function was improved in patients with epilepsy but little or no cognitive function change was found in short-term follow-up testing (6 months to 2 years) of pediatric epilepsy patients. A study by Puka and Smith²⁴ employed an extended postoperative follow-up time and increased sample size and reported a significant increase in intellectual function in the extended follow-up time. Therefore, a long follow-up observation time was used in this study of patients undergoing functional hemispherectomies. These patients were all children older than 6 years old or adults, which was more conducive to assessing changes in intelligence. The results showed that postoperative cognitive function significantly improved ($P = 0.006$), further emphasizing that the surgical technique was safe and worth promoting.

Other risk factors that may affect the development of postoperative intelligence also were analyzed. The results showed that the patients with epilepsy onset ages older 1 year (16/23) had greater

postoperative FSIQ scores ($P = 0.032$). Cognitive impairment occurs in the initial stage of epilepsy onset and becomes more serious over time. The earlier the onset of seizures, the more serious the impact on brain development and the later intelligence of the child.

In addition, postoperative FSIQ scores increased more in patients with no postoperative contralateral hemisphere discharge (12/23) after surgery ($P = 0.001$). The FSIQ scores in patients with postoperative epileptic seizures (16/23) were significantly greater than those in patients who were not seizure-free ($P < 0.001$), consistent with the results in other studies.^{11,12,15}

Moreover, there was a significant difference between the duration of the epilepsy course and increases in postoperative FSIQ scores ($P = 0.05$), suggesting that a long duration of preoperative epileptic seizures may affect the recovery of postoperative cognitive function.^{11,12,15} A shorter disease course duration has been reported to be beneficial to the recovery of postoperative language ability. Language proficiency detection is very important for intelligence assessment, so the duration of the preoperative course is likely to be a factor potentially influencing postoperative cognitive function. There was no statistical difference in FSIQ changes between children and adults ($P = 0.377$). The follow-up time in this paper was not associated with cognitive function changes ($P = 0.951$), and this result confirmed that the screened patients were all ones with a long follow-up time.

CONCLUSIONS

In summary, this study showed that successful functional hemispherectomy was a reliable means to effectively treat refractory epilepsy. Furthermore, it significantly improved the cognitive function of patients. Moreover, pathologic type, a significant reduction in postoperative epileptic seizures or seizure frequency, postoperative EEG normalization, and prolonged postoperative follow-up time were identified as essential conditions for cognitive improvement.

Limitations

Since the conclusions made in this paper were based on a small sample size, they may have limited generalizability. Additional data collection and expansion of the sample size are planned so that the data can be more deeply analyzed and more meaningful conclusions drawn. Due to the limited diagnostic ability of the pathology department in our hospital, the pathologic results could only be used as a reference for disease treatment and did not classify or analyze the pathologic types.

ACKNOWLEDGMENTS

We thank the patients and their families for the participation in this study. We thank Dr C. Wang for assisting in preparation of this manuscript.

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Conflict of interest statement: supported by the Project supported by the National Science Foundation for Young Scientists of China (Grant No. 81971206).

Received 17 September 2019; accepted 10 December 2019

Citation: *World Neurosurg.* (2020) 135:e520-e526.

<https://doi.org/10.1016/j.wneu.2019.12.058>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

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